XRPL Validator — Daily Engineering Report · 2026-09-06

Period: 2026-09-06 00:00 to 24:00 US Eastern (2026-09-06 04:00 to 2026-09-07 04:00 UTC); ledgers #106,792,708 to #106,815,061. Every count from the ledger and the validator's logs stops at #106,815,061. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 36 engine receipts recorded (20 fixed, 0 explained, 16 without a recorded fix).
- **Changes:** 0 deployments, 12 commits merged into main, 12 engine findings fixed (F185–F197).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#106,792,708–#106,815,061), by step:

Step	Ledgers with a receipt
Step 1 · native engine	36
Step 2 · records	0
Step 3 · order	0

Step	Ledgers with a receipt
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,793,245	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F185, commit a8e8cde, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-02	—	#106,796,893	Step 1 · native engine	1 object(s) differed in content	fixed: F186, commit 347380f, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-03	—	#106,796,939	Step 1 · native engine	1 object(s) differed in content	fixed: F187, commit 0b4ea2a, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-04	—	#106,800,767	Step 1 · native engine	1 object(s) differed in content	fixed: F188, commit 2502ad8, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-05	—	#106,800,826	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-06	—	#106,800,968	Step 1 · native engine	2 object(s) differed in content	fixed: F190, commit d9c5a54, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-07	—	#106,800,984	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 ledger change(s) the network made were absent from ours: A7B6454B (deleted), 25A867AD; 1 object(s) differed in content	fixed: F189, commit d923dee, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-08	—	#106,803,506	Step 1 · native engine	2 ledger change(s) the network made were absent from ours: EB2F18EA (deleted), DFA3B6DD (deleted); 2 object(s) differed in content	fixed: F191, commit fcfb7fc, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-09	—	#106,804,064	Step 1 · native engine	5 ledger change(s) the network made were absent from ours: 0D49784F, FEBBE8C4 (deleted), 82880E94; 15 object(s) differed in content	fixed: F192–F193, commit 6617922, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-10	—	#106,804,073	Step 1 · native engine	4 ledger change(s) the network made were absent from ours: 504E7705, D26CDCC8, F52A3ABA (deleted)	fixed: F192–F193, commit 6617922, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-11	—	#106,804,619	Step 1 · native engine	1 object(s) differed in content	fixed: F194, commit f562fff, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-12	—	#106,804,746	Step 1 · native engine	1 ledger change(s) the network made were absent from ours: 5F4DA292; 2 change(s) of ours the network did not make; 7 object(s) differed in content	fixed: F205, commit 2b27ffd, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-13	—	#106,806,079	Step 1 · native engine	5 object(s) differed in content	fixed: F197, commit dcf0734, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-14	—	#106,806,465	Step 1 · native	2 ledger change(s) the network made were absent from ours: DFA3B6DD (deleted), E77A4169 (deleted); 2 object(s)	fixed: F201, commit 7c49aa8, deployed in cycle 121 (2026-09-

ID	Time	Ledger(s)	Step	Description	Disposition
			engine	differed in content	11 17:39 UTC)
R-15	—	#106,807,323	Step 1 · native engine	1 object(s) differed in content	fixed: F200, commit a8439a9, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-16	—	#106,810,920	Step 1 · native engine	4 object(s) differed in content	fixed: F198, commit c1b1e51, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-17	—	#106,810,959	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-18	—	#106,811,005	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-19	—	#106,811,211	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-20	—	#106,811,321	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-21	—	#106,811,369	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-22	—	#106,811,381	Step 1 · native engine	12 object(s) differed in content	fixed: F203, commit 955de48, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-23	—	#106,811,410	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-24	—	#106,811,453	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-25	—	#106,811,549	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-26	—	#106,811,607	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-27	—	#106,811,643	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-28	—	#106,811,689	Step 1 · native engine	4 object(s) differed in content	no disposition recorded
R-29	—	#106,811,761	Step 1 · native engine	4 object(s) differed in content	fixed: F198, commit c1b1e51, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-30	—	#106,812,132	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-31	—	#106,813,010	Step 1 · native engine	6 change(s) of ours the network did not make	no disposition recorded
R-32	—	#106,813,390	Step 1 · native engine	1 object(s) differed in content	no disposition recorded

ID	Time	Ledger(s)	Step	Description	Disposition
R-33	—	#106,813,399	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-34	—	#106,813,814	Step 1 · native engine	4 object(s) differed in content	fixed: F204, commit 888ecec, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-35	—	#106,814,855	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 change(s) of ours the network did not make	fixed: F199, commit 79395eb, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-36	—	#106,814,864	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 change(s) of ours the network did not make	fixed: F199, commit 79395eb, deployed in cycle 121 (2026-09-11 17:39 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-06-01, INC-01 is INC-2026-09-06-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
04:38 UTC	xrpl-validator	`a8e8cde`	fix(xrpl-ledger): F185 — an NFTokenCreateOffer with a past Expiration is tecEXPIRED
08:40 UTC	xrpl-validator	`0b4ea2a`	fix(xrpl-ledger): F187 — a created DID object carries Flags: 0
08:40 UTC	xrpl-validator	`347380f`	fix(xrpl-ledger): F186 — an in-limited multi-path AMM slice truncates its out
12:43 UTC	xrpl-validator	`2502ad8`	fix(xrpl-ledger): F188 — TrustSet quality fields: side-specific, absent when zero
13:45 UTC	xrpl-validator	`d923dee`	fix(xrpl-ledger): F189 — AccountDelete's age rule is Sequence + 255 against the ledger being built
13:45 UTC	xrpl-validator	`d9c5a54`	fix(xrpl-ledger): F190 — an implied inter-gateway tail keeps a path's strand active
15:44 UTC	xrpl-validator	`fcfb7fc`	fix(xrpl-ledger): F191 — self-offers beyond the strict limit: the pool pre-empts the tip only from inside the limit
16:49 UTC	xrpl-validator	`6617922`	fix(xrpl-ledger): F192+F193 — XRP received in a flow does not fund the same flow

Merged	Repo	Commit	Subject
17:37 UTC	xrpl-validator	`f562fff`	fix(xrpl-ledger): F194 — a fee-less exhausting slice is debited the folded remainder
18:24 UTC	xrpl-validator	`7fb9c54`	fix(xrpl-ledger): F196 — CredentialCreate refuses a past Expiration; a mint's offer half expires like NFTokenCreateOffer's
03:36 UTC	xrpl-validator	`dcf0734`	fix(xrpl-ledger): F197 — a bridge walk's exhausting fill is bounded by the taker's line as the ledger carries it
03:40 UTC	xrpl-validator	`6d31efc`	chore: remove the Glow submission draft from the public tree

Findings fixed

- **fix(xrpl-ledger): F185 — an NFTokenCreateOffer with a past Expiration is tecEXPIRED** (`a8e8cde`, xrpl-validator) `hasExpired(ctx.view, ctx.tx[~sfExpiration])` — the parent ledger's close time at or past the offer's Expiration — is the FIRST test of NFTokenCreateOffer::preclaim (NFTokenCreateOffer.cpp), before the token is looked up. The native transactor had no expiration test and created the offer.
- **fix(xrpl-ledger): F187 — a created DID object carries Flags: 0** (`0b4ea2a`, xrpl-validator) `sfFlags` is soeREQUIRED on the DID ledger format, so rippled serialises `Flags: 0` (`22 00000000`) right after the entry type of every DID it creates, whether or not DIDSet touches it. Our DIDSet built the object without the field.
- **fix(xrpl-ledger): F186 — an in-limited multi-path AMM slice truncates its out** (`347380f`, xrpl-validator) `limitStepIn` hands the in-limited slice to `AMMOffer::limitIn(…, roundUp=false)`, which in multi-path mode is `Quality::ceilInStrict` → `divRoundStrict(limit, rate, false)`: the 17-digit integer quotient TRUNCATED to 16 digits (STAmount.cpp divRoundImpl, roundUp=false — no bump in muldivRound, canonicalizeRound down). Our slice sizing floored a ~17-digit quotient in `me_muldiv` and re-normalised to 16 digits at …
- **fix(xrpl-ledger): F188 — TrustSet quality fields: side-specific, absent when zero** (`2502ad8`, xrpl-validator) TrustSet::doApply writes the sender-side quality — `bHigh ? sfHighQualityIn : sfLowQualityIn` (and Out) — only when the transaction's value is non-zero and not QUALITY_ONE; zero (or 1e9) `makeFieldAbsent`s the field. Ours copied the transaction's `QualityIn`/`QualityOut` onto the line JSON under the TRANSACTION field names, so a RippleState went out carrying sfQualityIn (20) and sfQualityOut (21) — fields that do …
- **fix(xrpl-ledger): F189 — AccountDelete's age rule is Sequence + 255 against the ledger being built** (`d923dee`, xrpl-validator) `(*sleAccount)[sfSequence] + kSeqDelta > ctx.view.seq()` → tecTOO_SOON, with kSeqDelta = 255 and `view.seq()` the ledger under construction — the parent's sequence plus one (AccountDelete.cpp:287-289). Ours compared Sequence + 256 against the PARENT and answered tecNO_PERMISSION: two ledgers too strict, and the wrong code. `TxResult::TooSoon` ("tecTOO_SOON", fee-claiming) is added for it.
- **fix(xrpl-ledger): F190 — an implied inter-gateway tail keeps a path's strand active** (`d9c5a54`, xrpl-validator) toStrand's normalization appends the deliver issuer as an ACCOUNT element when the path ends holding the delivered currency under another issuer (PaySteps.cpp), so the strand closes with DirectSteps across the two gateways' mutual line — a real step, checked by DirectStepI::check (no line ⇒ the strand is not built). Our `mixed_layout` counted only the path's own account elements as "real" runs, so this implied tail …
- **fix(xrpl-ledger): F191 — self-offers beyond the strict limit: the pool pre-empts the tip only from inside the limit** (`fcfb7fc`, xrpl-validator) Two rules of the beyond-strict level walk in offer crossing, both from BookStep.cpp's stream loop: * The pool's turn (`tryAMM(offers.tip().quality())`) pre-empts the level's

CLOB offers only when its anchored offer is itself INSIDE the limit. Beyond it the offer fails the quality threshold, nothing is consumed, `execOffer` steps onto the tip, and a self-offer there is removed (`limitSelfCrossQuality`, "Remove this …

- **fix(xrpl-ledger): F192+F193 — XRP received in a flow does not fund the same flow** (`6617922`, xrpl-validator) Two findings from the #106804064–#106804073 receipts, both in offer.rs.
- **fix(xrpl-ledger): F194 — a fee-less exhausting slice is debited the folded remainder** (`f562fff`, xrpl-validator) rippled's in-limited fill debits the taker `stpAmt.in = remainingIn`, and `remainingIn` is SendMax less the ascending 16-digit fold of the saved per-iteration ins (StrandFlow.h). The walk's `rem_gets` already re-derives that fold at every iteration boundary, but the AMM turn's remaining GROSS cap — the figure an exhausting slice is debited verbatim (the in-limited gross-primary rule, #106455039) — came from the …
- **fix(xrpl-ledger): F196 — CredentialCreate refuses a past Expiration; a mint's offer half expires like NFTokenCreateOffer's** (`7fb9c54`, xrpl-validator) Two rippled 3.3.0 checks ported line-for-line, no mainnet specimen yet (both were listed as open no-specimen gaps after F182/F185): * CredentialCreate::doApply — `closeTime > *optExp` → tecEXPIRED, judged before the reserve check. STRICT: an Expiration equal to the parent close still creates. No `exp != 0` guard. * NFTokenMint::preclaim — `hasExpired(ctx.view, ctx.tx[~sfExpiration])` is the first test of the Amount …
- **fix(xrpl-ledger): F197 — a bridge walk's exhausting fill is bounded by the taker's line as the ledger carries it** (`dcf0734`, xrpl-validator) rippled's DirectStep limits an iteration's in by `maxSrcToDst`, read through PaymentSandbox::balanceHookIOU = min(live line balance, original − Σdebits). The exhausting slice's verbatim gross remainder (F50/F51: budget − the 16-digit fold of the spends) drifts an ulp from the sequential per-iteration line chain on a long crossing, and rippled takes the smaller. The direct walk had this bound (F122/F133/F169b); the …

Docs and operations

- **chore: remove the Glow submission draft from the public tree** (`6d31efc`, xrpl-validator)

Independence program

No independence record for UTC day 2026-09-06; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_3_0	2026-08-28 11:15 UTC	2026-09-11 11:15 UTC

Amendments enabled on mainnet: 93.

Open items

- Receipt R-05 (#106,800,826): open, under investigation.
- Receipt R-17 (#106,810,959): open, under investigation.
- Receipt R-18 (#106,811,005): open, under investigation.
- Receipt R-19 (#106,811,211): open, under investigation.
- Receipt R-20 (#106,811,321): open, under investigation.
- Receipt R-21 (#106,811,369): open, under investigation.
- Receipt R-23 (#106,811,410): open, under investigation.
- Receipt R-24 (#106,811,453): open, under investigation.
- Receipt R-25 (#106,811,549): open, under investigation.
- Receipt R-26 (#106,811,607): open, under investigation.
- Receipt R-27 (#106,811,643): open, under investigation.
- Receipt R-28 (#106,811,689): open, under investigation.
- Receipt R-30 (#106,812,132): open, under investigation.
- Receipt R-31 (#106,813,010): open, under investigation.
- Receipt R-32 (#106,813,390): open, under investigation.
- Receipt R-33 (#106,813,399): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Size-Weighted Owner Reserve Units** (fees) Add a per-LedgerEntryType weight table (a protocol constant: 1 unit for fixed-size objects like Offer/Escrow, ceil(serializedSize/192 bytes) for varia
- **MPT Issuer Holder-Cap Enforcement** (rwa) Add sfMaxHolderCount and sfHolderCount SFields to MPTokenIssuance plus a tfHolderCapped flag marking the issuance as cap-enforced. MPTokenAuthorize's
- **Policy-Scoped Pseudo-Account Session Keys** (accounts) Add a SessionKeySet transaction that creates an owner-reserved ltSESSION_KEY object (keylet::sessionKey(account, seq)) carrying sfSessionPublicKey, sf

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-06 04:38	09-06 04:38	xrpl-validator	`a8e8cde`	fix(xrpl-ledger): F185 — an NFTokenCreateOffer with a past Expiration is tecEXPIRED
09-06 08:40	09-06 08:40	xrpl-validator	`0b4ea2a`	fix(xrpl-ledger): F187 — a created DID object carries Flags: 0
09-06 08:40	09-06 08:40	xrpl-validator	`347380f`	fix(xrpl-ledger): F186 — an in-limited multi-path AMM slice truncates its out
09-06 12:43	09-06 12:43	xrpl-validator	`2502ad8`	fix(xrpl-ledger): F188 — TrustSet quality fields: side-specific, absent when zero
09-06 13:03	09-06 13:45	xrpl-validator	`d923dee`	fix(xrpl-ledger): F189 — AccountDelete's age rule is Sequence + 255 against the ledger being built
09-06 13:45	09-06 13:45	xrpl-validator	`d9c5a54`	fix(xrpl-ledger): F190 — an implied inter-gateway tail keeps a path's strand active
09-06 15:44	09-06 15:44	xrpl-validator	`fcfb7fc`	fix(xrpl-ledger): F191 — self-offers beyond the strict limit: the pool pre-empts the tip only from inside the limit
09-06 16:49	09-06 16:49	xrpl-validator	`6617922`	fix(xrpl-ledger): F192+F193 — XRP received in a flow does not fund the same flow
09-06 16:57	09-06 17:37	xrpl-validator	`f562fff`	fix(xrpl-ledger): F194 — a fee-less exhausting slice is debited the folded remainder
09-06 18:24	09-06 18:24	xrpl-validator	`7fb9c54`	fix(xrpl-ledger): F196 — CredentialCreate refuses a past Expiration; a mint's offer half expires like NFTokenCreateOffer's
09-07 03:36	09-07 03:36	xrpl-validator	`dcf0734`	fix(xrpl-ledger): F197 — a bridge walk's exhausting fill is bounded by the taker's line as the ledger carries it
09-07 03:40	09-07 03:40	xrpl-validator	`6d31efc`	chore: remove the Glow submission draft from the public tree

Times in UTC. A commit appears on the day it was committed to main or merged into it.